

Odocoileus virginianus Genome Exploration

Species: *Odocoileus virginianus*

NCBI Assembly Accession: GCF_023699985.2

Assembly Level: Scaffold

FASTA File: Odocoileus virginianus_genomic.fna.gz

Genome Source: NCBI

Approximate File Size: 770.1 MB

Objective

- Summarize basic genome assembly characteristics.
- Explain N50, L50, genome size, sequence count, and GC content.
- Analyze sequence length distribution.
- Apply a length filter and assess its effects on the assembly.
- Identify ORFs and distinguish them from confirmed genes.
- Document a reproducible Galaxy analysis in GitHub.

Tools Used

- Galaxy – genome analysis and data processing
- gfastats – calculation of genome assembly statistics
- Compute Sequence Length – determination of individual sequence lengths
- Histogram/Plotly – visualization of sequence-length distribution
- Filter Sequences by Length – removal of sequences shorter than 10 kb
- Filter Sequences by ID - used to select and extract specific genome sequences based on their sequence IDs.
- getorf – identification of possible open reading frames (ORFs)

Important Settings

- Complete genome FASTA was used for the initial assembly statistics.
- Sequence lengths were calculated for the genome sequences.
- A 10 kb minimum-length filter was applied as an educational filtering experiment.
- ORFs were identified using getorf.
- The ORF results were treated as potential coding regions rather than confirmed genes.

Key Results

- The original assembly contained 481 sequences with a total length of 2,423,263,331 bp. The maximum sequence length was 108,574,468 bp, with an N50 of approximately 66.6 Mb, an L50 of 15, and a GC content of 41.56%. The sequence-length distribution showed a mixture of long and short sequences, but the assembly was dominated by very long sequences. After applying the 10 kb filter, the number of sequences decreased from 481 to 371, while the total assembly length decreased only slightly to 2,422,532,368 bp. This indicates that the removed short sequences contributed relatively little to the total genome length, although they may still contain biological information.

Short Interpretation

- Overall, the *Odocoileus virginianus* assembly appears relatively contiguous based on the long sequences, N50, L50, and maximum sequence length. The GC content of 41.56% provides information about the genome's nucleotide composition. The ORF analysis demonstrated that computational methods can identify potential protein-coding regions, but ORFs cannot automatically be considered real genes because some may occur by chance and require additional evidence for confirmation.

Galaxy History

Galaxy History used for this analysis:

<https://galaxy-main.usegalaxy.org/histories/view?id=bbd44e69cb8906b50239c0db6d3d7ea9>